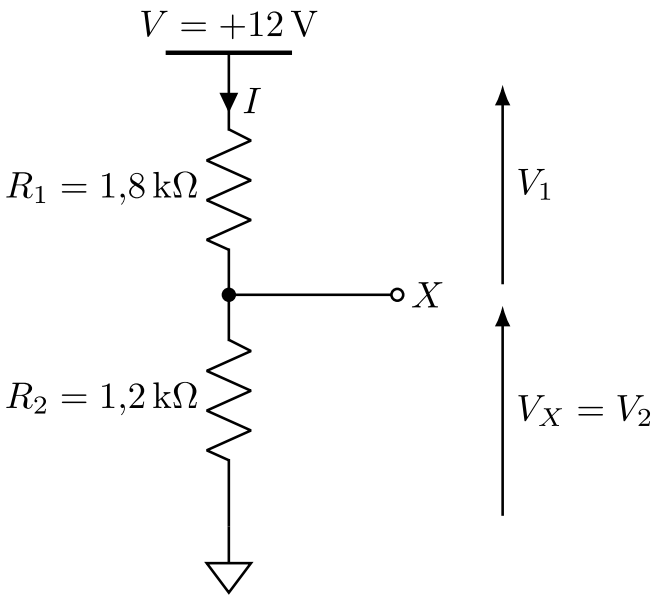
# Voltage and current dividers, power and energy

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## The voltage divider

A **voltage divider** is made of two resistances $R_1$ and $R_2$ connected **in series** between the supply voltage $V$ and ground. At the middle point $X$ we get a voltage $V_X$ **smaller** than $V$: the supply voltage is "divided" between the two resistances.




### Deriving the formula

- **1.** The two resistances are in **series**: the total resistance is $R_1 + R_2$.
- **2.** A single current flows in the circuit, found with Ohm's law: $I = V / (R_1 + R_2)$.
- **3.** The voltage at point $X$ is the voltage across $R_2$: $V_X = R_2 \cdot I$.
- **4.** Substituting the current gives the divider formula.

> **Voltage divider:** $V_X = V \cdot R_2 / (R_1 + R_2)$
>
> **Voltage across $R_1$:** $V_1 = V \cdot R_1 / (R_1 + R_2)$
>
> **Check:** $V_1 + V_X = V$

The voltage is shared **in proportion to the resistances**: the larger resistance takes the larger voltage. If $R_1 = R_2$, $V_X$ is exactly half of $V$.

**Careful:** the formula holds **with no load**, i.e. when nothing is connected to point $X$, or the connected load has a resistance much larger than $R_2$. A load connected at $X$ is in parallel

with $R_2$ and lowers $V_X$.

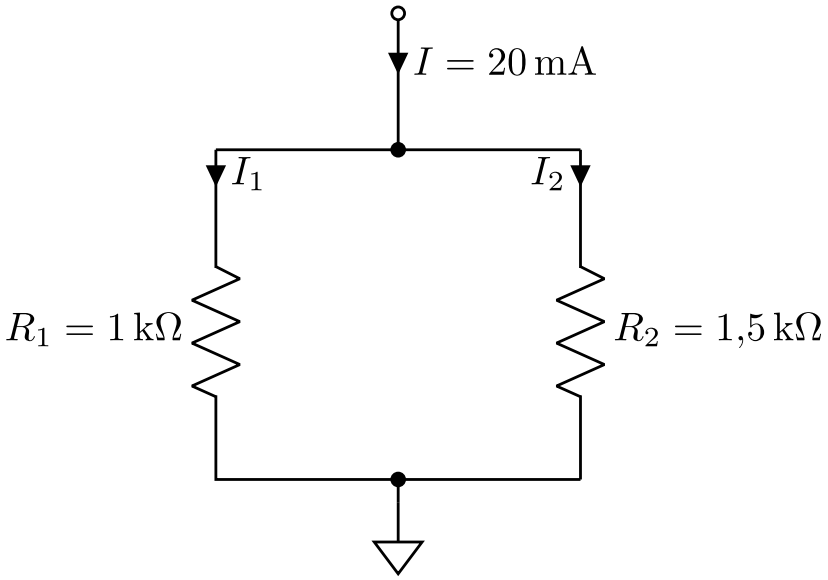
### Worked example

$V$ = 12 V, $R_1$ = 1.8 kΩ, $R_2$ = 1.2 kΩ (the circuit above).

- **total resistance:** $R_1 + R_2 = 1.8 + 1.2 = 3$ kΩ
- **current:** $I$ = 12 V / 3 kΩ = **4 mA**
- **voltage at *X*:** $V_X = 12 \cdot 1.2 / 3 =$ **4.8 V**
- **voltage across $R_1$:** $V_1 = 12 \cdot 1.8 / 3 =$ **7.2 V**
- **check:** 7.2 + 4.8 = 12 V ✓

## The current divider

A **current divider** is made of two resistances connected **in parallel**. The total current $I$ entering the node **splits** into the two branches: $I_1$ flows in $R_1$, $I_2$ flows in $R_2$.




### Deriving the formula

- **1.** The two resistances are in **parallel**: they have **the same voltage** $V$ across them.
- **2.** The equivalent resistance of the parallel is $R_{eq} = R_1 \cdot R_2 / (R_1 + R_2)$, so $V = R_{eq} \cdot I$.
- **3.** The current in each branch is $I_1 = V / R_1$ and $I_2 = V / R_2$.
- **4.** Substituting $V$ gives the divider formula.

> **Current in $R_1$:** $I_1 = I \cdot R_2 / (R_1 + R_2)$
>
> **Current in $R_2$:** $I_2 = I \cdot R_1 / (R_1 + R_2)$
>
> **Check:** $I_1 + I_2 = I$

**Watch the "crossed" resistances:** in the formula for $I_1$ the numerator contains $R_2$, and vice versa. Current takes the easier path: **the smaller resistance carries the larger**

**current**.

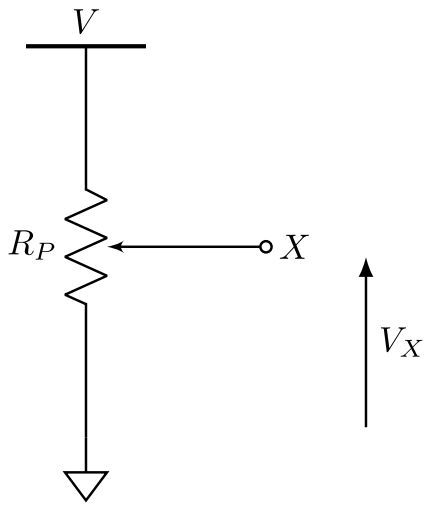
### Worked example

$I$ = 20 mA, $R_1$ = 1 kΩ, $R_2$ = 1.5 kΩ (the circuit above).

- **current in $R_1$:** $I_1 = 20 \cdot 1.5 / (1 + 1.5) = 20 \cdot 1.5 / 2.5 =$ **12 mA**
- **current in $R_2$:** $I_2 = 20 \cdot 1 / 2.5 =$ **8 mA**
- **check:** 12 + 8 = 20 mA ✓
- **voltage across the parallel:** $V = R_1 \cdot I_1 = 1$ kΩ $\cdot$ 12 mA = **12 V** (equal to $R_2 \cdot I_2 = 1.5$ kΩ $\cdot$ 8 mA)

## The potentiometer

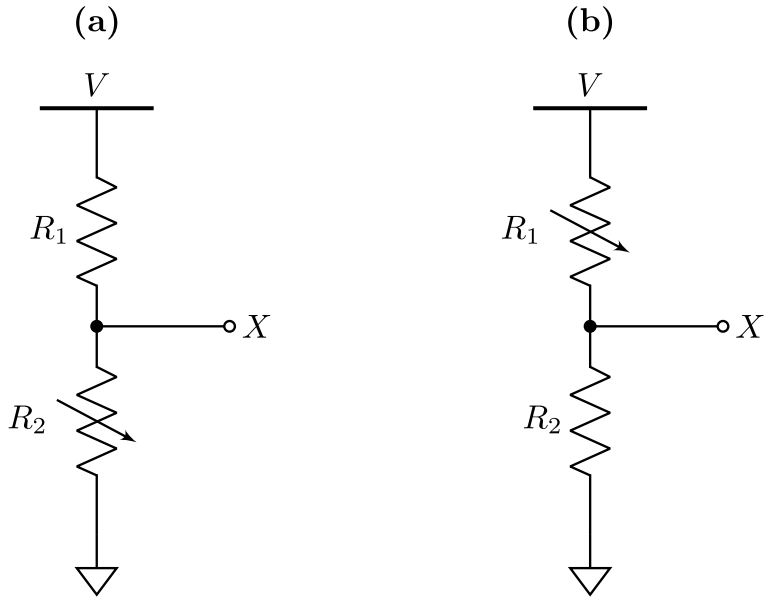
A **potentiometer** is a resistance with **three terminals**: the two ends and a **wiper** that slides along the resistive track. The wiper splits the total resistance $R_P$ into two parts: it is an **adjustable voltage divider**.




With the ends connected between $V$ and ground, turning the wiper changes $V_X$ smoothly **from 0 V to $V$**. Potentiometers are used to adjust volume, brightness, motor speed, a reference voltage.

## Where to put the potentiometer: instead of $R_1$ or $R_2$?

A **fixed resistance** is often used together with a potentiometer (or a variable resistance). The fixed resistance **limits the current** when the potentiometer is at zero and sets one of the two limits of the adjustment.




| Position | Minimum $V_X$ | Maximum $V_X$ | When to use it |
|---|---|---|---|
| **(a)** potentiometer instead of $R_2$ (towards ground) | 0 V | $V \cdot R_{2max} / (R_1 + R_{2max})$ | to adjust **from zero** up to a maximum value |
| **(b)** potentiometer instead of $R_1$ (towards $V$) | $V \cdot R_2 / (R_{1max} + R_2)$ | $V$ | to adjust from a minimum value **up to $V$** |

**Example (a).** $V$ = 12 V, fixed $R_1$ = 1 kΩ, $R_2$ potentiometer from 0 to 10 kΩ.

- with $R_2$ = 0: $V_X$ = **0 V**, and the current is at most 12 V / 1 kΩ = 12 mA, because $R_1$ limits it
- with $R_2$ = 10 kΩ: $V_X$ = 12 · 10 / 11 ≈ **10.9 V**
- the voltage is adjusted from 0 V to about 10.9 V

**Example (b).** $V$ = 12 V, $R_1$ potentiometer from 0 to 10 kΩ, fixed $R_2$ = 1 kΩ.

- with $R_1$ = 10 kΩ: $V_X$ = 12 · 1 / 11 ≈ **1.09 V**
- with $R_1$ = 0: $V_X$ = **12 V**, and the current is at most 12 V / 1 kΩ = 12 mA
- the voltage is adjusted from about 1.09 V to 12 V

**Rule of thumb:** put the potentiometer **on the side of the limit you want to reach**: towards ground (instead of $R_2$) to reach 0 V, towards the supply (instead of $R_1$) to reach $V$. The fixed resistance prevents a **short circuit** when the potentiometer is at zero.

## Electrical power

**Power** tells how much energy a component absorbs (or delivers) every second. In a resistance all the electrical energy turns into **heat** (Joule effect): that is why resistors get warm.

- Symbol: $P$ · unit: **watt** (W).
- Multiples and submultiples: 1 kW = 1000 W · 1 mW = 0.001 W.

> **Power:** $P = V \cdot I$
>
> **With Ohm's law:** $P = R \cdot I^2$ ; $P = V^2 / R$

Use the handiest form for the data: $V$ and $I$ known → $P = V \cdot I$; $R$ and $I$ known → $P = R \cdot I^2$; $V$ and $R$ known → $P = V^2 / R$.

### Rated power of resistors

Every resistor can dissipate at most a certain power, its **rated power**: the most common are **1/4 W** (0.25 W), 1/2 W and 1 W. If the calculated power exceeds it, the resistor overheats and may burn. It is wise to choose a resistor with a **margin**, for example at least twice the calculated power.

### Worked examples

- **Resistor at 12 V.** $R$ = 1 kΩ, $V$ = 12 V → $I$ = 12 mA; $P = V \cdot I = 12 \cdot 0.012 =$ **0.144 W** = 144 mW. A 1/4 W resistor is fine.
- **Resistor at the limit.** $R$ = 100 Ω, $I$ = 50 mA = 0.05 A → $P = R \cdot I^2 = 100 \cdot 0.05^2 =$ **0.25 W**: exactly the limit of a 1/4 W resistor, better use a 1/2 W one.
- **Electric heater.** $V$ = 230 V, $R$ = 26.45 Ω → $P = V^2 / R = 230^2 / 26.45 =$ **2000 W** = 2 kW.
- **Powers in the voltage divider** of the first example ($I$ = 4 mA): $P_1 = V_1 \cdot I = 7.2 \cdot 0.004$ = 28.8 mW; $P_2 = V_X \cdot I = 4.8 \cdot 0.004$ = 19.2 mW; total $P = V \cdot I = 12 \cdot 0.004 =$ **48 mW** = 28.8 + 19.2 ✓.

## Electrical energy

**Energy** is the power absorbed over the whole operating time: the longer an appliance stays on, the more energy it uses.

> **Energy:** $E = P \cdot t$

- With power in **watts** (W) and time in **hours** (h), energy is in **watt-hours** (Wh).
- Multiple: 1 **kWh** (kilowatt-hour) = 1000 Wh. It is the unit shown on the electricity bill.
- Time always goes in **hours**: 30 min = 0.5 h · 15 min = 0.25 h · 10 min ≈ 0.167 h.

**A note on the joule.** The SI unit of energy is the **joule** (J): 1 J = 1 W · 1 s, i.e. with power in watts and time in seconds. The joule is a very small unit: 1 Wh = 3600 J. That is why **watt-hours** and **kilowatt-hours** are used for electrical systems and appliances.

### Worked examples

- **Resistor left on.** The 0.144 W resistor of the previous example stays on for 10 h: $E = 0.144 \cdot 10 =$ **1.44 Wh**.
- **Hair dryer.** $P = 1200$ W for 15 min = 0.25 h: $E = 1200 \cdot 0.25 =$ **300 Wh** = 0.3 kWh.
- **LED bulb.** $P = 10$ W, on 5 h a day for 30 days: $t = 5 \cdot 30 = 150$ h; $E = 10 \cdot 150 = 1500$ Wh = **1.5 kWh**.
- **Heater and cost.** $P = 2000$ W for 3 h: $E = 2000 \cdot 3 = 6000$ Wh = **6 kWh**. At a price of €0.25/kWh the cost is $6 \cdot 0.25 =$ **€1.50**.

## Formula summary

> **Voltage divider:** $V_X = V \cdot R_2 / (R_1 + R_2)$
>
> **Current divider:** $I_1 = I \cdot R_2 / (R_1 + R_2)$ ; $I_2 = I \cdot R_1 / (R_1 + R_2)$
>
> **Power:** $P = V \cdot I$ ; $P = R \cdot I^2$ ; $P = V^2 / R$
>
> **Energy:** $E = P \cdot t$ (W · h = Wh) ; 1 kWh = 1000 Wh ; 1 Wh = 3600 J